Introduction to data organisation

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Slides: http://bit.ly/eglen2019-data (CC-BY license)

Acknowledgements

Data parasites everywhere

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Rule 1: Data should be shared

Given the cost of generating data, I think data relating to a publication should be shared along with the paper.

Funders (and increasingly many, but not all, journals) agree.

The Lancet

Longo DL, Drazen JM (2016) Data Sharing. N Engl J Med 374:276–277 http://dx.doi.org/10.1056/NEJMe1516564.

Data sharing sounds great … but

1. Someone not involved in collecting data may misunderstand it.

2. "a new class of research person will emerge --- people who had nothing to do with the design and execution of the study but use another group's data for their own ends, possibly stealing from the research productivity planned by the data gatherers, or even use the data to try to disprove what the original investigators had posited. There is concern among some front-line researchers that the system will be taken over by what some researchers have characterized as **research parasites''**."

Response from the community was to embrace the term "resesarch parasite", e.g. https://twitter.com/dataparasite

This is just good science. I think there are few valid reasons for not sharing data.

Why share your data?

Biggest beneficiary might be your future self. See also "Selfish reasons" in next talk.

Writing a good data management plan (DMP)

FAIR data standards

Box 2 | The FAIR Guiding Principles

To be Findable:
F1. (meta)data are assigned a globally unique and persistent identifier
F2. data are described with rich metadata (defined by R1 below)
F3. metadata clearly and explicitly include the identifier of the data it describes
F4. (meta)data are registered or indexed in a searchable resource

To be Accessible:
A1. (meta)data are retrievable by their identifier using a standardized communications protocol
A1.1 the protocol is open, free, and universally implementable
A1.2 the protocol allows for an authentication and authorization procedure, where necessary
A2. metadata are accessible, even when the data are no longer available

To be Interoperable:
I1. (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
I2. (meta)data use vocabularies that follow FAIR principles
I3. (meta)data include qualified references to other (meta)data

To be Reusable:
R1. meta(data) are richly described with a plurality of accurate and relevant attributes
R1.1. (meta)data are released with a clear and accessible data usage license
R1.2. (meta)data are associated with detailed provenance
R1.3. (meta)data meet domain-relevant community standards

https://www.nature.com/articles/sdata201618

Things to be wary of

GDPR, Ethics, anonymisation.

Getting help

CambridgeData Champions (phd students, postdocs, faculty)
https://www.data.cam.ac.uk/data-champions-search

Office for Scholarly Communication

a model adopted elsewhere

The Turing Way

Many key findings in publications are either not independently verified, or fail verification when it is attempted (Baker, 2016).

Never mind the headline fraud cases…

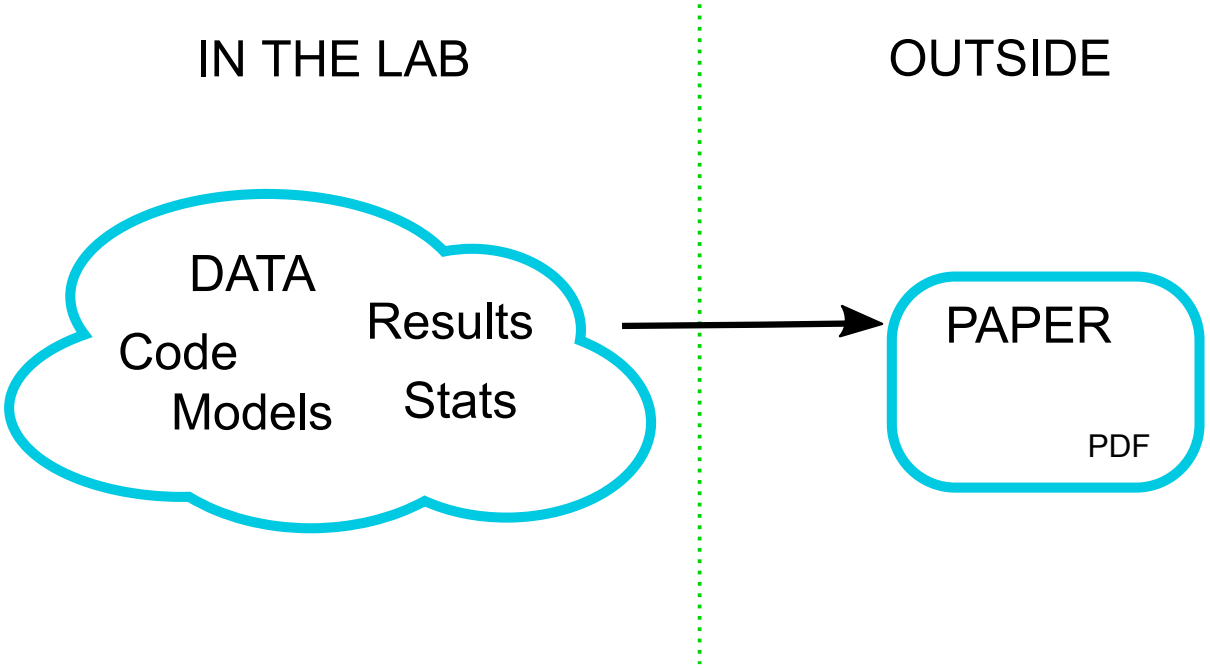
Data goes missing quite quickly (Vines et al 2014).

A systematic study examining 300 papers published in 2016/17 in Journal of Computational Physics, *a journal that promotes sharing of digital artefacts*, found that only 5.6% made artefacts available (Stodden et al 2018).

Who's got the time to verify results?

Disclaimer: do I mean "reproducibility" or "replicability" (Barba 2018)?

Science as an inverse problem



Reproducible papers

- "Reproducible papers" are slowly gaining in popularity. People providing data and code.
- Journals are exploring systems like Code Ocean and BinderHubs.
- These systems are good, and when they work are super. BUT…

Difficulties with current approaches

- Who pays for mybinder instances?
- "Reproducible forever" is a BIG challenge. My first attempt lasted about 6 months. Dockerhub for waverepo. Is "forever" achievable/useful?
- Interactive approach suitable for shorter jobs. But how about for long jobs (days--months)?
- Do editors / reviewers have time to check that code and data work? I can recall only one comment (positive!) about running my code by a reviewer.
- Pushing the bar up and up is good for quality, but how practical is it? What is "good enough?"

What is "good enough"?

Barnes (2010) convinced me that something is better than nothing.

Published online 13 October 2010 | *Nature* **467**, 753 (2010) | doi:10.1038/467753a

Column: World View

Publish your computer code: it is good enough

Freely provided working code — whatever its quality — improves programming and enables others to engage with your research, says Nick Barnes.

Nick Barnes

"All models [**papers/programs**] are wrong but some are useful"

Not everyone agrees: Milkowski et al 2018 suggest that paper should contain only key material.

Our thoughts

- Having code and data archived with paper already a win.
- Dockerfile helps you understand the environment, even if it's broke.
- Having an independent assessment that the code ran *once* in an indepedent environment is useful knowledge for a reviewer.
- So how about lowering the bar instead of raising it?
- Proposal to Wellcome Open Research Fund Summer 2018 declined, but brought Daniel and Stephen together.

CODE CHECK

- http://www.codecheck.org.uk
- CODE CHECK will be a system for generating "Certificates of Computation"
- These state that code was run on a given date on a given platform and generated particular results (figures, tables).
- Output log files are archived with the results.
- Relevant outputs archived on zenodo with permanent URL.
- Aim. Paper includes a CODE CHECK badge to highlight "Code/data available with archive of independent verification of outputs."
- This is not checking that code is scientifically correct -- simply that code runs, and show what was generated.

Example certificate

https://sje30.github.io/codecheck/eglen2016/eglen2016-crc.html

How much should be shared?

- What is practical to ask people to share?
- We follow the suggestion from modelDB that at least one key figure / table / result should be replicable.
- Toy examples welcome; long jobs might be too difficult.
- Who checks the figures? No-one. It stands alone. Reviewers could scan them.

Challenges to tackle

- How automated can we get? repo2docker tool adoption.
- How much load will this add for journals?
- Provide training examples for people to adopt / case studies.
- Compute infrastructure for long jobs (can be handled locally for now).
- Running commercial software (handled locally).

Target journals to work with

- Neurons, Behaviour, Data, Theory
- Gigascience
- ELife

Dates

https://xkcd.com/1179/